

Question	Answer	Marks	Guidance
4(a)	correct circuit symbol used for cell with internal resistance	B1	Allow just a cell symbol, without the resistor. Ignore presence or absence of dotted box around it. not a battery symbol
	correct symbol for LDR connected to terminals of cell	B1	
	correct symbol for voltmeter connected in parallel with LDR	B1	If symbol for LDR incorrect, allow voltmeter connected in parallel with component drawn, provided only one component drawn.
4(b)(i)	zero error / calibration error	B1	Allow 'contact resistance error in the connections'. Allow 'resistance of connecting wires' Allow 'not calibrated' BOD Apply List Rule
4(b)(ii)	$P = V^2 / R$ or $P = VI$ <u>and</u> $V = IR$ or $P = I^2 R$ <u>and</u> $V = IR$ $= 4.78^2 / 230$ or $= 4.78 \times (4.78 / 230)$ or $= (4.78 / 230)^2 \times 230$	C1	
	$= 0.099 \text{ W}$	A1	
4(b)(iii)	$E = V + Ir$ $r = (4.90 - 4.78) \times 230 / 4.78$ or $4.78 / 4.90 = 230 / (230 + r)$ or $(4.90 - 4.78) / 4.90 = r / (230 + r)$ or $(4.90 - 4.78) / 4.78 = r / 230$ or $4.90 / (4.78 / 230) = r + 230$	C1	Allow $E = I(R + r)$
	internal resistance = 5.8Ω	A1	

Question	Answer				Marks	Guidance
4(b)(iv)		decreases	unchanged	increases	B1	Allow symbols other than ticks if no ticks seen
	resistance of LDR			✓		
	current in cell	✓			B1	
	p.d. across LDR			✓	B1	